Race, explain and transfer

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare your predicted result with measured performance. Use the results to explain your design choices.

SUCCESS CRITERIA - I can...

1. Report measured distance, time and average speed.
2. Compare actual speed with a stated prediction.
3. Use measurements or observations to explain three physical design choices.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “relative deviation” and “energy budget” to explain your answer. Sentence starters are optional.

relative deviation: difference from prediction compared with its magnitude

energy budget: an account of all energy stored and transferred

revision: a change made after checking evidence or reasoning

01 Warm-up: bring back an earlier idea

A **60**-tooth driver turns a **480**-tooth gear. **Give ideal output**/input torque and speed ratios and say what you treated as true about energy losses.

Word help: torque = the turning effect of a force about an axis

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

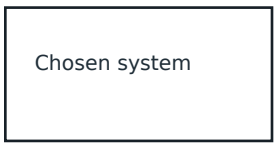
Reference: average speed = distance/time
relative deviation = |measured - predicted|/|predicted| × **100%**.
Prediction must be nonzero. **Record actual track**, starting method, and setup; this compares with a prediction, not symmetric percent difference.

Worked reasoning: A car covers **3 m** in **2 s**: average speed **1.5 m/s**. A complete explanation also uses its circuit, gearing, friction and measurements from repeated runs; one race time does not isolate a cause.

Your turn: Draw an energy-flow diagram from battery to motion and other transfers; annotate it with one actual result, clearly labeled as measured.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw or annotate your final** car.
2. **Label three choices involving wheels**, gears, alignment, wiring or mass, then connect each to a physics idea.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

04 Apply the physics

A supplied car covers **6.0 m** in **8.0 s**; its predicted speed was **0.90 m/s**. **Complete the table**, find measured speed and percent deviation from prediction. **Give a possible explanation** and **name a measurement** that could test it.

Sentence starter: I predict … . I used … to make this prediction because … .

Distance (m)	Time (s)	Measured v	Predicted v
___	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Use your actual race record to defend one successful design choice and **name one revision** for a future build. Cite data or a specific observation for each.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A motor is **64%** efficient and the following generator is **72%** efficient. **Find the combined useful fraction** in sequence.
Label input, useful output and other energy transfers.

Word help: generator = a device converting mechanical input to electrical output

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

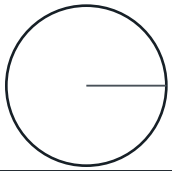
Marzano check - circle: 1 2 3

07 Exit check

Transfer: a similar car uses twice the wheel diameter with the same loaded wheel rpm. **Predict ideal speed change**, then **explain why testing** is still necessary.

Word help: ideal = using stated simplifying conditions

Sentence starter: I predict … . I used … to make this prediction because … .



Radius / diameter: ____

Turns per second: ____

Travel per turn: ____ Ground speed: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Supplied input is **6 V×1 A** for **4 s**. A **0.5 kg** car reaches **2 m/s** on a level track from rest.
1. **Find input energy** and final translational kinetic energy.
2. **Explain why their ratio alone** is not total drivetrain efficiency.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3